Personalised Email Recommendation System

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***Abstract*—This paper enlightens the importance of unsubscribing to unwanted emails and establishes a pipeline involving combination of ensemble machine learning techniques coupled with Natural language processing. The model built on publicly available enron dataset was used for benchmarking purposes. This model was then used to label the synthetic dataset created by the authors of this paper. Once the synthetic emails have been labelled, the metadata of these emails are then passed to an Artificial Neural Network which was trained to classify emails as important and unimportant. The user will then be prompted to unsubscribe or delete these emails. This helps in mitigating carbon emissions caused due to emails.**

***Index Terms*—Email, unsubscribe, Natural Language Processing, Artificial Neural Network**

I. Introduction

Achieving environmental sustainability is an integral goal these days owing to multiple factors one of which is majorly being climate change. Many organizations have proposed to achieve net zero carbon emission in a bid to achieve environmental sustainability. One of the steps involved in achieving environmental stability includes reducing carbon footprint. There are multiple sources that leave behind carbon footprint. Few important sectors that contribute to considerable amount of carbon footprint are Information & Communication Technology, Travel, Food, Fashion, Water, Waste, Energy. In order to determine the amount of carbon footprint they leave behind a method called as Life Cycle Assessment (LCA) has been adopted. LCA estimates the environmental impacts of a product or service. This article [1] elaborates on the steps involved in LCA and the benefiters of LCA.

Information and Communication technology (ICT) sector generally consists of user devices, Access networks, Data centres and enterprise networks, operator activities. The authors of this research [2] dives deep into the impact of ICT from the year 2010 to 2015. They claim that the ICT sector's carbon footprint in the year 2015 was about 730 Mt of CO_2 and that this corresponds to 1.4% of the global carbon footprint. According to the authors, it was observed that user devices account for the largest share of carbon footprint among others in the year 2015. They also briefly discuss the expectations of carbon footprint wherein they suspect a linear growth in the number of subscription and an exponential growth of IoT.

The carbon footprint left by emails also falls under the category of ICT. Research from Trinity College of Dublin [3] shows that on an average a person receives an equivalent of 1,652 g of CO_2 of email which is approximately 0.6 tonnes of CO_2. This paper focuses on mitigating carbon footprint left by emails by building a personalized email recommender system which enables users in deciding to unsubscribe to unwanted newsletters and subscriptions. This act can help saving approximately 0.03g of CO_2 to 26g of CO_2 depending on the size of the emails, number of recipients, number of attachments and other factors.

Section II consists of due diligence conducted in order to decide on the optimal algorithm or combination of algorithms that can be employed to build an optimal solution. Section III discusses the methodology employed and the step carried out while building the model. Once the model was built and tested, the model was evaluated using certain standard evaluation metrics to judge the performance of the model. These evaluation metrics are discussed in section IV. This paper also discusses potential future work that can be carried out to enhance the performance and add new features in section V.

The artefacts with respect to this project has been maintained in a github repository.

II. Related Work

The main idea behind this study [1] is to find how effective the pre-trained embedding models can be used for the email classification along with the deep learning models like Convolutional neural networks(CNN) and Long short term

[1] https://ecochain.com/knowledge/life-cycle-assessment-lca-guide/

[2] https://www.ericsson.com/en/reports-and-papers/research-papers/the-future-carbon-footprint-of-the-ict-and-em-sectors

[3] https://www.tcd.ie/itservices/news/emails-carbon-footprint/

memory(LSTM). In this paper embedding models like BERT and GLOVE have been used both are pre-trained models that can be used to determine the relationship between the words present in the dataset. Also, Machine learning models such as support vector machine (SVM), random forest (RF), logistic regression (LR), Gaussian, Naive Bayes (GNB), Multinomial Naive Bayes (MNB), and AdaBoost The dataset used in this project is SpamAssassin and Enron. Two types of experiments have been conducted in this project one is by applying the deep learning algorithms with the BERT and GLOVE model and the second one is to apply them without the BERT and GLOVE model. The experiment output says that the SVM model outperforms all machine learning models with 98.70% accuracies in the SpamAssassin dataset and 98.50% accuracy in the Enron dataset. The LSTM and CNN model along with the BERT embedding technique gives the accuracy of 98.21% and 98.44% on the SpamAssassin dataset and 98.70 and 98.50% on the Enron dataset. The LSTM and CNN along with the GLOVE embedding technique achieve the accuracy of 98.75% and 98.59%, respectively, on the SpamAssassin dataset and 98.86% and 99.06% on the Enron dataset. The results show that the CNN model on the Glove embedding performs better than the other models.

This study [2] aims at studying the Email document sentiment classification. In comparing to the data present in the social media like Twitter and review data. Email data has several different characteristics like varying lengths and sentiment indicators. To analyze these types of data the author has introduced a novel approach called Sentiment Sequence encoding incorporating CNN(SSE-CNN). The Sentiment Sequence Encoding is built on the Long short term memory (LSTM) with tree-based sentiment position features along with sentence weighing. A sentence-level dependency matrix with the concatenation of a scaled SWN lexical feature matrix is used to represent the email document. Three datasets have been used in this British Columbia Dataset (BC), EnronFFP dataset, and Personal Archive. The SSE-CNN has achieved the accuracy of 88.6%, 74.3%, and 82.1% on these datasets respectively.

The author in this study [3] has used several machine learning algorithms like Naive Bayes, SVM, and KNN. Several preprocessing techniques like Tokenization, Stop-words removals, Converting into lowercase, and Stemming has been applied. The word embedding techniques like word2vec and TF-IDF have been used The result shows that SVM with word2vec performed better than the other machine learning models in email classification.

This paper [4] deals with classifying the document based on their types and clustering them for decision making. Both the real and artificial types of datasets like NEWS 20, Reuters, emails, and research papers have been used in this project. The word embedding technique like the Term frequency-inverse Document frequency has been used along with Fuzzy K-means and hierarchical clustering algorithm. The experiment on this dataset has been conducted in two different phases in the phase one the fuzzy k means and HAC algorithms have been applied in the small part of this dataset based on these results which algorithm gives the best values those algorithms are chosen and applied on the large portion of these datasets. For NEWS 20, Reuter dataset Hierarchical clustering algorithms give a lesser entropy score and a higher F-score, for the research paper dataset Fuzzy K means gives the higher F1-Score and lesser entropy. For the email dataset, there is not much difference so either one can be applied to the large dataset. In phase two, the HAC model has been applied on the large data set and the classification is obtained.

In this paper [5], the author has used the Long Short Term Memory (LSTM) with stratified sampling, a deep learning model, for classifying the emails into four different types of categories like fraudulent, suspicious, harassment, and normal. The dataset used in this study is ”SeFACED: Semantic-based Forensic Analysis and Classification of E-Mail Data. Since the dataset is imbalanced the oversampling technique called Synthetic Minority Oversampling Technique (SMOTE), has been used to balance the dataset. The balance dataset achieves an accuracy of 97% and the imbalanced dataset achieves an accuracy of 80%.

The author [6] uses the LSTM along with the word embedding technique Glove. The eight models in the LSTM model by fine-tuning the parameters. The eight models of the LSTM can be obtained by applying different combinations of optimizers and hidden layers. In doing so the sixth model with the optimizer Adam and Hidden layer Tanh gives an accuracy of 95.17% along with applying the Glove model.

In this study [7], the author has introduced a new model called Learn#. It contains four different types of models Student models, Reinforcement learning models, Teacher models, a discriminator models. The student models extract the text features. The RL model filters the predicted results by the students. The teacher model again checks and reclassifies the results predicted by the student model. If the student model reached its threshold it will be eliminated to improve the stability and the training speed of the module.

Sentiment classification is an important task in understanding the meaning of the text. BERT is one of the Natural language processing techniques which can be used in text classification. The results show that BERT outperforms the other deep learning models like recursive, recurrent, and convolutional neural networks [8], in classifying the Sentiment in the Stanford Sentiment Treebank(SST) Dataset.

In this paper [9] BERT (Bidirectional Encoder Representations from Transformers) model which is a Natural Language processing model, has been implemented for detecting the spam or ham. It use the attention model for finding the context of the text. Even though the machine learning and deep learning models have been used for email classification the NLP model enhances the accuracy in finding the Spam or Ham. The results of this module have been compared with DNN(DEEP Neural Network) and a BiLSTM (bidirectional Long Short Term Memory) layer and two stacked Dense layers. The Results show that the Bert transformer model achieves an accuracy of 98.67% and an F1-Score of 98.66% which is better than the

Bi-LSTM model.

In this paper [10], the combination of Support Vector Machine(SVM) and LDA (Latent Dirichlet Allocation)is used for text classification. In this work, Practical Swarm optimization (PSO) is proposed for optimizing error penalty parameter C in SVM to improve text classification and is termed LDA-SVM-.PSO The dataset used in this work was the Sogou News Chinese classification corpus data. The model was compared with the other two models LDA-KNN and LDA-SVM. The LDA-KNN showed the least accuracy compared to the other two algorithms. By using PSO accuracy was increased by 3.95%, precision rate by 1.19%, recall by 6.08%, and F-Measure by 3.89%. Although the text classification was increased by obtaining the PSO optimization technique with LDA-SVM but the dataset was limited to only chine texts so the future work suggested to use this algorithms to other text messages.

Another paper [11], has proposed text document clustering by k-means along with the dimension reductional technique. Initially pre-processing step was applied to remove noise and calculated the TF-IDF score weights Later dimensional reduction techniques Singular Vector Decomposition (SVD) and Non-negative Matrix Factorization (NMF) were applied to these weights in a final step the reduced feature vectors are given to the k-means clustering algorithm. The cluster performance was evaluated by accuracy, and it was found that SVD k-means performed well compared to NMF k-means. These dimensional reduction techniques can be considered to apply in our work to improve clustering efficiency.

The goal of this work [12] was to apply Deep Learning to the subject classification. In this project, two feature selection methods were applied initially, and found the word2doc method was more stable later these word vectors were given to LSTM for binary classification. This LSTM was compared with CNN and LSTM had achieved a better precision value compared to CNN.

The purpose of this work was to develop a spam detection system that could handle both textual and voice-assisted e-mail messages. Genetic Decision Tree Processing with Natural Language Processing (GDTPNLP) is the name of this email spam detection approach. Natural Language processing is inbuilt into this algorithm. In terms of classification accuracy, speech-to-text conversion accuracy, and other metrics, the GDTPNLP algorithm delivered excellent results. GDTPNLP is a hybrid method that combines genetic and decision tree concepts with Natural Language Processing. The Principal Component Analysis scheme dynamically collects features to reduce data overhead in the classification process. Despite the method's high performance, the purpose is to classify spam, which includes voice mail data. This level of sophistication is not required for the dataset in question. Some of the data pre-processing steps from the paper are used in this study [13].

III. Methodology

The Cross-Industry Standard Process for Data Mining(CRISP-DM) is used to achieve the study's goal of creating a pipeline for identifying important and non-important data. This section explains how CRISP-DM was applied to address the research question.

A. Business Understanding

Carbon footprints in the Information and Communication technology (ICT) sector has been a cause of concern. Researchers estimate that the ICT sector's carbon footprint is approximately 730 Mt CO_2-equivalents or 1.4% of overall global emissions in the year 2018, and the sector uses 800 TWh or 3.6% of the global electricity for its operation (based on 2015 data). Major contributors to carbon footprints in the ICT sector are Storage facilities (Data Centres), Emails, Wireless communication devices which include Moblie phones, laptops, modems, etc. The goal of this paper is to attempt to reduce carbon footprints left by emails. As established earlier, it is estimated that an individual email leaves approximately 0.03g CO2e to 26g CO2e of carbon footprint. Building a personalized email unsubscription recommendation system can help in reducing the inflow of a lot emails which in turn reduces the carbon footprint considerably.

B. Data Understanding

In order to build an email recommendation system which classifies emails as important and unimportant, it is required to establish a metric which helps in the decision making of classifying emails as important and unimportant. For the purpose of this paper, Topic modelling is performed, and the emails are labelled important and unimportant manually as there is no global standard followed. As part of future work, a labelled data set can be used to improve on results.

C. Data Preparation

The data used for this project was obtained in two parts. The Enron email dataset which is publicly available was sampled and was used to calibrate the topic modelling and clustering models. These models help in classifying the emails as important and unimportant.

The second part includes a synthetic email dataset which has been created by the authors of this paper. The synthetic email dataset created contains emails from various subscriptions which include newsletter subscriptions, product subscriptions, etc. Apart from subscriptions, publicly available standard email templates are used to populate the inbox using an auto email generation script which selects a random template from a list of templates. The meta data of these emails are later extracted using i-map tools. The metadata that are retrieved from the mailbox are Email size, read status, attachment details and flags regarding the subscription, flagged and encryption status. The final classification is performed on this metadata.

The Pre-processing performed on the Enron dataset includes:

1) Handling missing values: The Enron dataset had a lot of null values. As part of the preprocessing, these null values were removed. Further investigation revealed that the synthetically generated data from Gmail contained some empty

email bodies. Emails like this were likewise removed from the database.

2) Feature selection: Only the body of the email was utilized for Clustering. The other available columns were dropped. Once the labeling is done a deep learning technique is used to build a classifier. The ID field and the cluster classes were among the 17 columns in the Gmail metadata used to train artificial neural networks. Date, sender, and subject were not used in training, thus they were removed.

3) Tokenize words: Each sentence was converted to list of words by eliminating punctuations and extra letters. The method simple_preprocess() from Gensim was ideal for this and was set to $deacc$=$True$ to eliminate the punctuation marks.

4) Removing stop words: Stopwords in English were retrieved from the nltk package, and additional words such as "from," "subject," "re," "edu," and "use" were added to the list. Clustering was not considered for any of these words and was removed by method called remove_stopwords()

5) Data lemmatization: Only nouns, adjectives, verbs, and adverbs were maintained, with the remainder being ignored. This aids in the elimination of non-relevant words for clustering algorithms.

6) Creating Bigram and Trigram : A bigram is the two adjacency words occurring together and Trigram is the three adjacency words occurring together. The phrases model from Gensim can create Bigrams and Trigrams. Min count and threshold are the two most significant parameters for Phrases. The greater the values of these parameters, the more difficult it is to combine words. In our project min_count was set to 5 and the threshold to 100.

7) Creating word embeddings: Word embedding techniques are used to represent words in number. These are also called as word vectorizers. Some of the common word embedding techniques used are TF-IDF, GloVe, Sentence-Transformers, FastText, CountVectorizers, etc. For the purpose of this project three different techniques have been followed.

a) TF-IDF: TF-IDF stands for term frequency inverse document frequency. TF-IDF gives a measure that takes the importance of word into consideration depending on how frequently a word appears in a document or a corpus. As the name suggests, TF-IDF is the ratio of frequency of a word in a document to the measure of how common a particular word is across all the documents in the corpus.

b) GloVe: GloVe stands for Global Vectors for word representation. It is an unsupervised learning algorithm for vectorizing words. GloVe uses Euclidean distance or Cosine similarity between words plotted on a vector space thereby quantifying relatedness of similar words and retaining the context of the word usage. It is developed by Jeffery Pennington, Richard Socher, Christopher D. Manning of Stanford university and is open source[4].

For the purpose of this paper, the pre-trained vectors of GloVe was used as benchmark and an ANN was run which vectorizes the entire corpus of the Enron Email dataset.

[4]https://nlp.stanford.edu/projects/glove/

c) SentenceTransformers: A sentence transformer named $all - MiniLM - L6 - v2$ is used for BERTopic modeling. Paragraphs and sentences are mapped into a 384-dimensional dense vector space that can be used for semantic search and clustering by this transformer.

8) Creating Dictionary and Corpus: The corpus and dictionary(id2word) are the two basic inputs to the LDA topic model. A unique ID is created in the document for each word using Genism. The mapping of (word id, word frequency) in a corpus is shown in Figure 1.

```
[[(0, 1), (1, 1), (2, 1), (3, 1), (4, 1), (5, 2), (6, 1), (7, 1), (8, 1), (9, 1), (10, 1), (11, 1), (12, 1)]]
```

Fig. 1. Mapping of word id and word frequency in a corpus

D. Modelling

1) Clustering of emails:

a) Latent Dirichlet Allocation: Latent Dirichlet Allocation (LDA) is a topic modeling algorithm implemented using Python's Gensim Package. It is used to extract topics from large text data. A topic is a representative of dominant keywords. LDA takes a proportional approach to topic modeling, treating each document as a collection of topics. And each topic is a collection of keywords, proportionately distributed. Once the number of topics is provided to the algorithm, it simply rearranges the distribution of topics in a document and achieves desirable topic keyword distribution. Along with the corpus and dictionary-created values the number of topics required was also given as input to the LDA model.The other important input parameters are -

- $chunk_size$ – the number of documents used in each training set and was set to 100
- $update_every$ - specifies the frequency with which the model parameters shall be updated and was set to 1
- $passes$ – the total training passes and was set to 100.
- num_topics was set to 5.

```
lda_model = gensim.models.ldamodel.LdaModel(corpus=corpus,
                                            id2word=id2word,
                                            num_topics=5,
                                            random_state=100,
                                            update_every=1,
                                            chunksize=100,
                                            passes=10,
                                            alpha='auto',
                                            per_word_topics=True)
```

Fig. 2. LDA model initialization

Figure 2 shows the initialization of LDA model. The LDA was built with 5 different topics and each topic was associated with a combination of keywords and this keyword was associated with a certain weight for a topic. The weightage importance of keyword was given by lda_model.print_topics() and output is shown in Figure 3.

The interpretation of Figure 3 for instance Topic 0 $(0, '0.029 * "league" + 0.024 \times "thegistsport" +$

```
[(0, '0.029*"league" + 0.024*"thegistsport" + 0.022*"ect" + 0.015*"season" + 0.015*"gist" + 0.014*"inserted_link" + 0.014*"nd_a
rticle" + 0.014*"rd_article" + 0.014*"st_article" + 0.013*"team"'), (1, '0.024*"https" + 0.024*"fce" + 0.013*"cubw_ybmluz" + 0.
010*"chm_ly" + 0.008*"say" + 0.007*"year" + 0.007*"news" + 0.005*"bv_zb" + 0.005*"story" + 0.005*"bv_tzwrpdw"'), (2, '0.014*"li
nk" + 0.012*"thank" + 0.011*"email" + 0.010*"know" + 0.009*"signature" + 0.008*"name" + 0.008*"let" + 0.008*"time" + 0.007*"ad
d" + 0.007*"get"'), (3, '0.542*"com" + 0.020*"courses_master" + 0.020*"list_utm_campaign" + 0.018*"utm_medium_email" + 0.013*"a
o_course" + 0.009*"master_list" + 0.007*"mc_cid_mc_eid" + 0.007*"course_tripdiscount" + 0.006*"enron" + 0.006*"utm_campaign_cou
rse"'), (4, '0.080*"style" + 0.036*"px" + 0.032*"table" + 0.032*"font_size" + 0.030*"www_atlasobscura" + 0.026*"tr" + 0.026*"em
ail_utm_term" + 0.025*"width" + 0.021*"text" + 0.020*"div"')]
```

Fig. 3. Importance of keywords in a topic

$0.022 \times$ "ect" $+ 0.015 \times$ "$season$" $+ 0.015 \times$ "$gist$" $+ 0.014 \times$ "$inserted_link$" $+ 0.014 \times$ "$nd_article$" $+ 0.014 \times$ "$rd_article$" $+ 0.014 \times$ "$st_article$" $+ 0.013 \times$ "$team$"')
Figure 3 describes top 10 keywords belonging to Topic 0 and the topics are league, thegistsport, ect, season, gist , inserted_link , nd_article , rd_article , st_article and team.The perplexity value of corpus was found as -8.8203.
LDA provides a method $get_document_topics()$ to retrieve the likelihood of a document falling under a topic by passing the corpus of words as arguments once the model has been constructed. The algorithm was fed a corpus of emails from the Gmail dataset, and labelling was done accordingly. Each email was assigned to the topic with the highest probability.

b) BERTopic: BERTopic assists with the creation of dense clusters that are easy to understand while also including relevant keywords in the topic descriptions. To do this, this strategy utilizes transformers such as sentence transformers and a class based variation of TF-IDF. This method will create the topics as well as the corresponding probability. A pre-trained language model converts each email body to its embedding representation. Because this data is primarily English, BERTopic is performed with a model that only supports English. The dimensionality of the generated embeddings is then lowered before clustering to improve the clustering process [14]. Finally, topic representations are retrieved from document clusters using a customized class-based variant of TF-IDF. The input parameters give to the BERTopic model are as follows:

- $language$ parameter is set to English as this dataset only contains data in English.
- $embedding_model$ used in this BERTopic modeling is $all - MiniLM - L6 - v2$ as mentioned earlier.
- nr_topics corresponding to number of topics is set to 5.
- $calculate_probabilities$ is set to True to calculate the probability distribution of each email falling under a topic.

The data's outliers are all labelled as belonging to topic -1 as can be seen from Figure 4. The BERTopic model also makes it possible to determine which topic a word belongs to. The required number of topics can be set for the BERTopic model and in this scenario, it is set to 5 same as in previous models. When the number of topics was reduced, the problem with BERTopic was that there were a lot of outliers. The model discovered 100+ topics with better separation if topic count limitations were not set. This number of topics did not correspond to the proposed pipeline. The probability distribution for each email falling under a topic was evaluated to overcome the problem of outliers. The email was assigned to the topic with the highest probability among them. The five

	Topic	Count	Name
0	-1	8869	-1_com_enron_https_px
1	0	1056	0_message_original_get_know
2	1	777	1_enron_com_dec_john
3	2	697	2_name_thank_product_link
4	3	592	3_post_link_add_name
5	4	333	4_magnet_lead_name_link

Fig. 4. Topics generated using BERTopic model

topics were then analyzed and classified as important or not important based on the words found in each.

c) K-Means: K-means clustering falls under unsupervised learning which is run on unlabelled data. It is used to identify groups in the data with the number of groups being 'K'. The algorithm assigns each of the data points in the data set to one of the K groups based on the features provided.

```
In [11]: clusters[0]

Out[11]: ['nn',
          'nikkei',
          'demographics',
          'underlying',
          'nyt',
          'genes',
          'immune',
          'inventories',
          'logic',
          'losers',
          'calculated',
          'flexible',
          'narrative',
          'liability',
          'default',
          'sin',
          'semiconductor',
          'excuse',
          'transparency',
```

Fig. 5. Sample K-means cluster

For this paper, After the pre-processing is carried out on the enron dataset, the data set has been tokenized into words. Then the most common words were identified and also duplicate words were removed. These words were vectorized using the pre-trained GloVe vectorizer. These word embeddings are then used to generate a cluster size of 5

2) Classification of emails:

a) Artificial Neural Network: Artificial Neural networks (ANN) are feed forward neural networks that consists of an input layer, hidden layers and an output layer. All the nodes of

the ANN are fully connected and the number of hidden layers determine how deep the neural network is. For this project, a simple ANN with 4 layers was used. The layer used for building the ANN model was the dense layer. The dense layer is one of the most commonly used layers that receives its input from previous neuron inputs and passes it to the next layer.

```
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 dense_50 (Dense)            (None, 7)                 91

 dense_51 (Dense)            (None, 6)                 48

 dense_52 (Dense)            (None, 7)                 49

 dense_53 (Dense)            (None, 1)                 8

=================================================================
Total params: 196
Trainable params: 196
Non-trainable params: 0
_________________________________________________________________
```

Fig. 6. Keras Sequential Model

Figure 6 shows the Keras sequential model used for the email classification that uses four dense layers. Only few layers were added as the number of emails in the synthetic dataset was less and to prevent overfitting. The activation function used for the denser layers was Rectified Linear unit (ReLu). The output function used was a dense layer with a sigmoid activation function that gives the output in terms of probability between 0 to 1. The model was compiled using Adam optimizer and "binary_crossentropy" as loss function as this is a binary classification model.

For training the model the data frame from the metadata csv file was split with the independent variables and the dependent variable stored separately. The subject, from and the date of the emails were removed from the input data frame. The values in the input data frame were then standardized using the StandardScaler() function. The input and the output data were then split using the "train_test_split" function with a test size of 0.2. The input train and test data frames were then converted into NumPy array for processing in the neural network.

The model was then fit with the input and the output values with a validation split of 0.2 which allocates 20% of the data for evaluation. The epoch is the number of cycles the entire dataset will pass through while the batch size is the number of values of data that would pass through at once. The epoch and the batch size chosen for this model was 10.

b) TensorFlow Decision Forests (TF-DF): Since the model uses a synthetic dataset, to further evaluate its performance TensorFlow Decision Forests was used. Decision Forests are algorithms that are made up of multiple decision trees like Random Forest, Gradient boosted Decision Tree[5]. The TensorFlow Decision (TF-DF) forests was created for training and understanding these Decision Forest models. The TF-DF model is only available in Linux and macOS and for hence for this project Google Colab Notebook was used.

[5]https://towardsdatascience.com/tensorflow-decision-forests-train-your-favorite-tree-based-models-using-keras-875d05a441f

The TF-DF model does not require any pre-processing of the data and can handle both categorical and numerical values. The Boolean values in the data frame were converted into integers of 0 and 1. They were divided for training and testing using the "train_test_split" with a test size of 0.2. For using the decision forests models the input in the pandas data frame were converted into TensorFlow datasets. The target variable "isImportant" was mentioned while converting and the task is mentioned as classification.

For the Decision Forest model, RandomForest was chosen due to its ability to work better in higher dimensions. The model was then compiled on the basis of accuracy and then the training data was fitted. The addiditon of sys_pipes helps in displaying the training logs and the hyperparameter used for training the model.

E. Process Flow

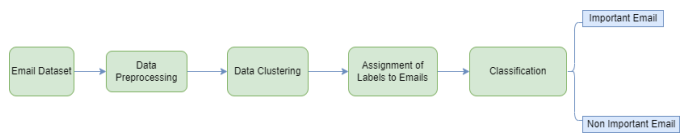

Fig. 7. Process Flow

Figure 7 shows the process flow of the proposed method. It includes clustering and classification processes. The email dataset was built by combining the Enron email dataset with Synthetic email data that was partially generated using a scheduler. After the data was merged, a clustering algorithm was used to label the unlabeled data. This labelled data was then associated with its metadata and used to train the classification algorithm.

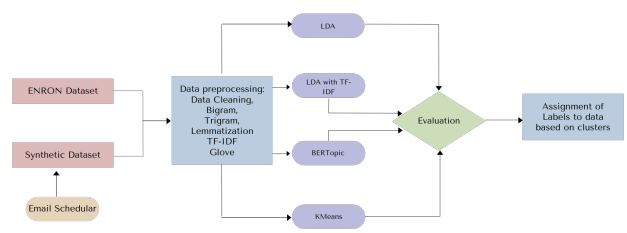

Fig. 8. Process flow of Clustering

Figure 8 shows the process flow of the clustering technique used in this work. As previously stated, the dataset was created, and clustering was performed using four different methods. LDA, LDA with TF-IDF, BERTopic, and KMeans are the models. Each clustering technique was assessed, and the best one was selected. The labels were assigned once the clustering was completed.

Figure 9 shows the process flow of the classification step involved. Training and Testing data are segregated from the labelled data. The training data was used to train the Neural

Network model for binary classification. The outcomes of the evaluation were captured using the testing data.

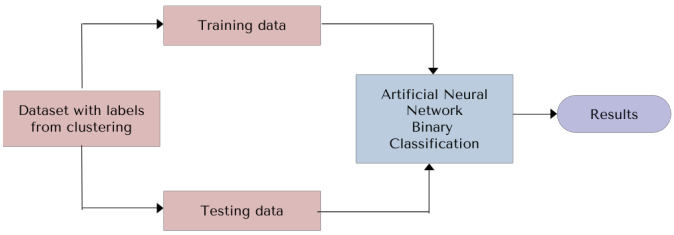

Fig. 9. Process flow of Classification

IV. Evaluation

A. LDA

The perplexity value and coherence score of LDA methods were examined. The results of LDA with and without TF-IDF were found to be identical. The topic labels were then added to the synthetic data, and it was discovered that the majority of the emails were linked to topic 2 as shown in Figure 11. This amount of class imbalance is not ideal for this work.

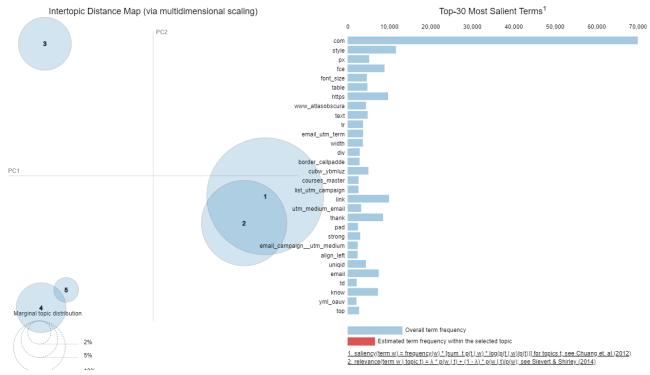

Fig. 10. Distribution of topics using LDA model

The distribution of topics and their keywords are shown in Figure 10. It can be seen that there is huge amount of irrelevant words coming under the topics. This is not ideal and indicates that the model is not clustering well.
The sensitivity and specificity values for this data were poor in a test run with a Neural Network created for binary classification. This discovery, when combined with the confusion matrix, revealed that the topics with the fewest values are misclassified. Modifying the data at hand to handle the class imbalance could be one technique to solve this problem. This was overlooked because other clustering approaches were being considered.

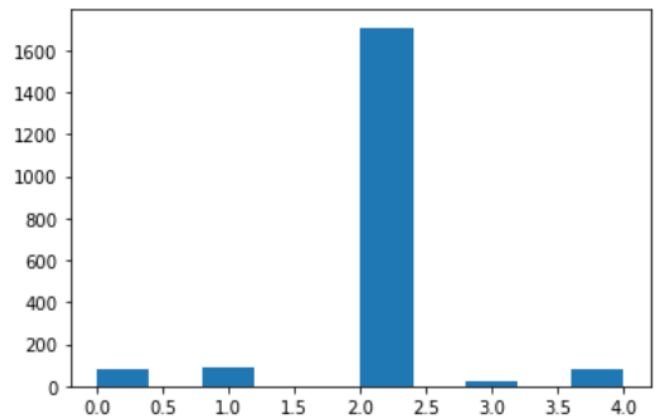

Fig. 11. Distribution of topics associated with synthetic data from LDA

B. BERTopic

Figure 12 shows the all the topics identified by BERTopic model and their top 5 words. When compared to the topics generated using LDA, it is clear that these topics make better sense.
There was a considerable difference in the topics generated,

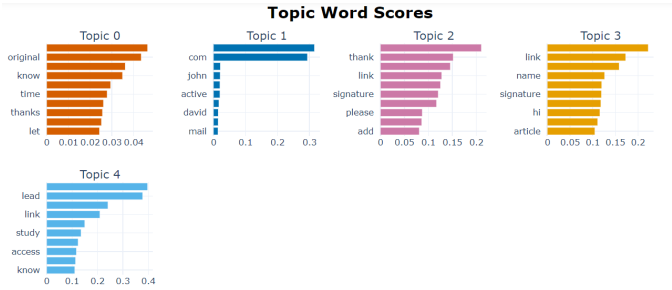

Fig. 12. Topic word scores of topics obtained from BERTopic

and there was no class imbalance in the distribution of subjects for the synthetic data. Hence BERTopic model was selected for clustering the unlabelled synthetic data.
After BERTopic modelling was used to generate the topics, a detailed study of the words of each topic revealed that Topic 1 was not important. Another topic was deemed unimportant since it had certain unnecessary words in the context of an email. The remaining topics were all deemed important. The message IDs and their corresponding topics were recorded into a CSV file so that the relationship between metadata could be determined using the message-id before the classification algorithm is trained.

C. K-Means

For evaluating the K means model, few of the standard metrics include Silhouette Coefficient, the Calinski Harabasz index, the Davies Bouldin index and the within the sum of squared error (WCSS). The values for these evaluation metrics was calculated as shown in Figure 13.
It is observed that the Silhouette coefficient is around 0.019

```
In [39]: silhouette_score(enrond_dataframe, labels)
Out[39]: 0.019130303

In [34]: calinski_harabasz_score(enrond_dataframe, labels)
Out[34]: 676.6872376406299

In [35]: davies_bouldin_score(enrond_dataframe, labels)
Out[35]: 5.955475306248163

In [36]: kmeans.inertia_
Out[36]: 1516227.0
```

Fig. 13. Evaluation Metrics - KMeans

which is too low. The silhouette coefficient is expected to be closer to +1 for a good performing model. Also, the Davies Bouldin index is also too high at around 5.95. The lesser the value of Davies Bouldin index the better the model performance. The WCSS also is observed to be very high and for a good model the number has to be on the lower side.

Owing to the observations made in these evaluation metric, it can be concluded that the K-Means model has a below average performance and can be rejected.

D. ANN

The ANN model was evaluated using the model.evaluate() function. The accuracy of the model was 86% and the total loss was 39.33%. The Fig shows the plot between the accuracy and validation accuracy of the model and the plot between the loss and the validation loss.

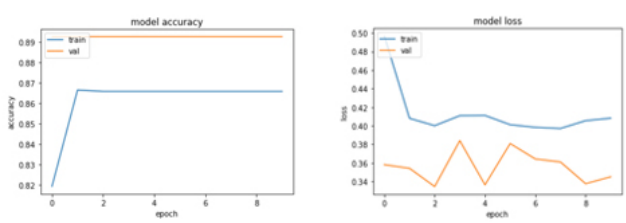

Fig. 14. Model Accuracy and Model loss of ANN

To completely evaluate the model, the model was used to predict the test dataset. The model outputs the probability of the email being important. A median threshold value was chosen to filter the probability to classify them as True and False. The predicted output was then compared with the actual output and the confusion matrix was visualized as shown in the Figure 15.

The model had a true positive rate of 70% and true negative rate of 17%. The model had a sensitivity of 99% which indicates that the model avoids classifying important emails as non-important. The model had erroneously classified 12% of the non-important emails as important. While this needs to be reduced, the classification of important emails will have a higher importance as it might result in the loss of critical information. The overall accuracy of this neural network model

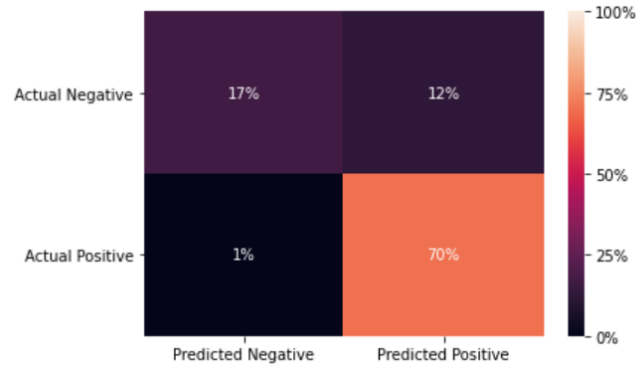

Fig. 15. Confusion Matrix - ANN

was 86%. While this is fairly high, the dataset used here is a synthetic dataset and hence the results may or may not the reflect the actual capability of the model.

	precision	recall	f1-score	support
0	0.96	0.58	0.72	116
1	0.85	0.99	0.91	280
accuracy			0.87	396
macro avg	0.90	0.78	0.82	396
weighted avg	0.88	0.87	0.86	396

Fig. 16. Evaluation Metrics - ANN

E. TensorFlow Decision Forests

To evaluate the TF-DF model, accuracy of the model has been chosen as the evaluation metric. The major advantage of TF-DF model over the traditional Decision Forests models is that they provide some functionalities for better interpretability. The fig 16 shows a single decision tree from the model with the depth of 4.

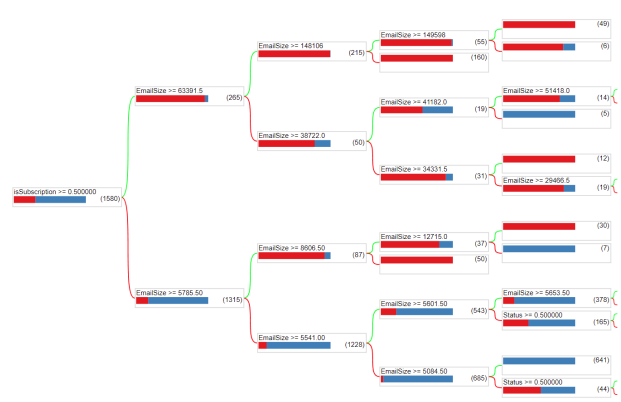

Fig. 17. TensorFlow Decision Forest Tree

The first 4 levels shows the classification of the first tree. The first split had been done on the isSubscription variable

which shows that most of the subscribed emails have been classified as not important as they are most probably newsletters and advertisements. To further delve into the understanding of the important features of the model, the minimal depth of all the variables is obtained as shown in the Figure 18.

```
{'MEAN_MIN_DEPTH': [("NoOfToValues" (1; #4), 8.190117240437939),
  ("isBCC" (1; #6), 8.190117240437939),
  ("isEncrypted" (1; #8), 8.190117240437939),
  ("isFlagged" (1; #9), 8.190117240437939),
  ("__LABEL" (4; #12), 8.190117240437939),
  ("AttachmentCount" (1; #0), 8.018655187350838),
  ("AttachmentSize" (1; #1), 8.000835502393015),
  ("Status" (1; #5), 7.938307126182083),
  ("isCC" (1; #7), 7.437517273318682),
  ("NoOfCCValues" (1; #3), 7.322792202391669),
  ("isPrimaryRecipient" (1; #10), 6.963634329744246),
  ("isSubscription" (1; #11), 5.880703790529286),
  ("EmailSize" (1; #2), 0.36713410275759556)],
```

Fig. 18. Minimal Depth of all variables

The minimal depth of a feature indicates the depth at which the observations are split and the closest one to the root of the tree. A low value for the minimal depth indicates a higher importance of the variable. Based on the synthetic dataset the three most important variables for classifying the emails were EmailSize, isSubscribtion which indicates if the mail is from subscription list and the isPrimaryRecipient which indicates if the mail is directly sent to the user. For further testing the model test data was used to evaluate the performance of the model. The confusion matrix is visualised as shown in the Figure 19.

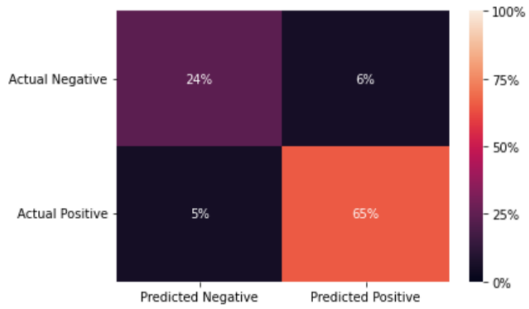

Fig. 19. Confusion Matrix - TFDF

The model was able to classify 65% of the true positive and the 24% of True negative values. The sensitivity of the model was 93% which was comparatively less than the ANN but the model has a higher specificity which indicates that it has classified less non important emails as important. The overall accuracy of the model was 89% for this synthetic dataset.

V. Conclusion and Future Work

The objective of this paper was to build a personalized email recommender system using email meta data. The purpose of this recommender system was to help users in deciding on which emails can they possibly unsubscribe to and delete. The idea behind this is to highlight the importance of an individual and the role they play in the journey of an organization to achieve environmental sustainability by contributing towards reducing carbon footprint.

The authors of this paper were able to build a prototype of the recommender system which was based on a synthetic dataset. As mentioned earlier, these results obtained may or may not reflect the capability of the system as they were not built on using real world dataset.

In this study, two datasets were used. The first was generated synthetically with the help of a scheduler, while the second was the Enron email data. The unlabeled data was clustered and labels were assigned to related metadata information using a clustering method. Following that, the data was used to train a classification algorithm.

Initially, for clustering the LDA model with and without the TF-IDF was used, and it was discovered that the majority of the emails were clustered into a single cluster. In terms of performance statistics, the K-Means model performed poorly. When BERTopic was used for clustering, it produced good clusters when compared to others. Because BERTopic outperformed all other clustering algorithms, it was chosen to label the synthetic data.

In this investigation, two classification approaches were applied. The first was a binary classification Neural Network. It had an 86% overall accuracy. The second classification algorithm used was Tensorflow Decision Forests. It had an 89% overall accuracy. The second one had a lower sensitivity than ANN but a higher specificity value.

As part of future work, a self-training algorithm can be built and provided as a web extension or as a plug-in. This self-training algorithm will replace the need to of the model to be calibrated using a third-party dataset and also this allows the recommendation system to be more personalized.

In terms of improving the current structure in place, dimensionality reduction technique can be used on the feature vectors to get feasible vector values. The research can also be carried forward to increase the accuracy by creating a better word corpus.

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